CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 28: Distance Formula

Topic-wise short notes with solved examples

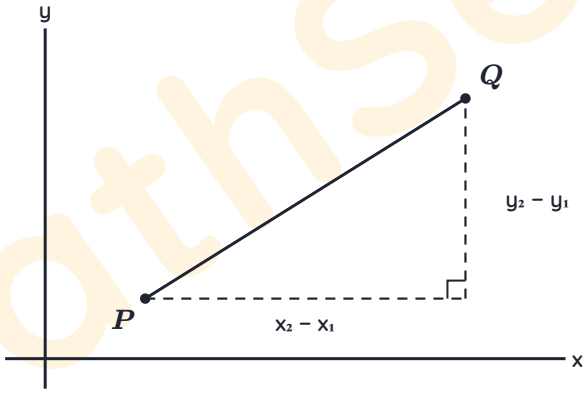
The formula for the distance between two points in the Cartesian plane, and its use to test for collinear points, identify triangles and quadrilaterals, and find points and circles.

1 The formula

- The distance between $P(x_1, y_1)$ and $Q(x_2, y_2)$ is

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- It comes from Pythagoras theorem: the horizontal step is $x_2 - x_1$ and the vertical step is $y_2 - y_1$.
- The distance of $P(x, y)$ from the **origin** is $\sqrt{x^2 + y^2}$.
- The order of the points does not matter, because the differences are squared.



PQ is the hypotenuse of a right triangle

Example 1.1 Find the distance between $A(2, 3)$ and $B(6, 6)$.

SOLUTION

$\sqrt{(6-2)^2 + (6-3)^2} = \sqrt{16 + 9}$

5 units

Example 1.2 Find the distance between $P(-3, 4)$ and $Q(5, -2)$.

SOLUTION

$\sqrt{8^2 + (-6)^2} = \sqrt{100}$

10 units

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Example 1.3 Find the distance of $(-5, 12)$ from the origin.

SOLUTION

13 units

2 Points on the axes

- Distance between $(x_1, 0)$ and $(x_2, 0)$ on the x-axis: $|x_2 - x_1|$.
- Distance between $(0, y_1)$ and $(0, y_2)$ on the y-axis: $|y_2 - y_1|$.
- Distance of (x, y) from the x-axis is $|y|$, and from the y-axis is $|x|$.

3 Finding an unknown co-ordinate

- Write the distance formula, square both sides and solve. There are often **two** answers.

Example 3.1 The distance between $A(2, -3)$ and $B(10, y)$ is 10 units. Find y.

SOLUTION

$64 + (y + 3)^2 = 100$, so $(y + 3)^2 = 36$ and $y + 3 = \pm 6$.

$y = 3$ or $y = -9$

Example 3.2 Find the point on the x-axis equidistant from $A(2, -5)$ and $B(-2, 9)$.

SOLUTION

Let the point be $P(x, 0)$. $PA^2 = PB^2$: $(x - 2)^2 + 25 = (x + 2)^2 + 81$.

$-4x + 29 = 4x + 85$, so $8x = -56$.

$(-7, 0)$

Example 3.3 Find the point on the y-axis equidistant from $(6, 5)$ and $(-4, 3)$.

SOLUTION

Let it be $(0, y)$: $36 + (y - 5)^2 = 16 + (y - 3)^2$, so $61 - 10y = 25 - 6y$.

$(0, 9)$

Example 3.4 Find a relation between x and y if $P(x, y)$ is equidistant from $A(3, 6)$ and $B(-3, 4)$.

SOLUTION

$(x - 3)^2 + (y - 6)^2 = (x + 3)^2 + (y - 4)^2$

$-6x - 12y + 45 = 6x - 8y + 25$

$3x + y = 5$

4 Collinear points

- Three points A, B, C are **collinear** if the sum of two of the distances equals the third: for example $AB + BC = AC$.

Example 4.1 Show that $A(1, 2)$, $B(3, 4)$ and $C(6, 7)$ are collinear.

SOLUTION

$AB = \sqrt{8} = 2\sqrt{2}$, $BC = \sqrt{18} = 3\sqrt{2}$, $AC = \sqrt{50} = 5\sqrt{2}$.

$AB + BC = AC$, **so the points are collinear**

5 Identifying triangles

Triangle	Test
Scalene	All three sides different
Isosceles	Two sides equal
Equilateral	All three sides equal
Right-angled	(longest side)^2 = sum of the squares of the other two
Right-angled isosceles	Two sides equal and Pythagoras holds

- Work with the **squares** of the lengths to avoid square roots.

Example 5.1 Show that $A(3, 0)$, $B(6, 4)$ and $C(-1, 3)$ are the vertices of a right-angled isosceles triangle.

SOLUTION

$AB^2 = 9 + 16 = 25$, $AC^2 = 16 + 9 = 25$, $BC^2 = 49 + 1 = 50$.

$AB = AC$ **and** $AB^2 + AC^2 = BC^2$**, so the triangle is isosceles with a right angle at A**

Example 5.2 Show that $(0, 0)$, $(2, 0)$ and $(1, \sqrt{3})$ form an equilateral triangle.

SOLUTION

The three squared distances are 4, $1 + 3 = 4$ and $1 + 3 = 4$.

All sides are 2 units

6 Identifying quadrilaterals

- Find the four sides and the two diagonals.

Quadrilateral	Sides	Diagonals
Parallelogram	Opposite sides equal	Not equal
Rectangle	Opposite sides equal	Equal
Rhombus	All sides equal	Not equal
Square	All sides equal	Equal

Example 6.1 Show that $A(1,1)$, $B(4,1)$, $C(4,4)$ and $D(1,4)$ are the vertices of a square.

SOLUTION

$AB = BC = CD = DA = 3$. $AC^2 = 9 + 9 = 18$ and $BD^2 = 9 + 9 = 18$.

All sides are equal and the diagonals are equal: a square

Example 6.2 Show that $P(-2,-1)$, $Q(1,0)$, $R(4,3)$ and $S(1,2)$ form a parallelogram.

SOLUTION

$PQ^2 = 9 + 1 = 10$, $RS^2 = 9 + 1 = 10$; $QR^2 = 9 + 9 = 18$, $SP^2 = 9 + 9 = 18$.

Opposite sides are equal, so PQRS is a parallelogram

7 Circles

- Every point on a circle is at distance r from the centre. To check whether a point lies on a circle, compare its distance from the centre with the radius.
- The **circumcentre** of a triangle is equidistant from its three vertices.

Example 7.1 A circle has centre $(2,1)$ and passes through $(5,5)$. Find its radius. Does $(-1,5)$ lie on it?

SOLUTION

$r = \sqrt{9+16} = 5$. Distance of $(-1,5)$ from the centre $= \sqrt{9+16} = 5$.

Radius 5; yes

Example 7.2 Find the centre of the circle through $(0,0)$, $(6,0)$ and $(0,8)$.

SOLUTION

Let the centre be (h,k): $h^2 + k^2 = (h-6)^2 + k^2$ gives $h = 3$; $h^2 + k^2 = h^2 + (k-8)^2$ gives $k = 4$.

Centre $(3,4)$; radius 5

8 Common mistakes

- Subtracting a negative co-ordinate wrongly: $5 - (-3) = 8$, not 2.
- Forgetting the square root, or forgetting to square the differences.
- Giving only one value when the equation gives two.
- Calling a quadrilateral a square after checking only the sides. A rhombus also has four equal sides; check the diagonals.
- Testing collinearity with two distances only.
- Writing a point on the x-axis as $(0, x)$.

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